

Métodos Matemáticos de la Física II

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Bla bla bla... resumen...

I. METRIC SPACE

A set X is called a *metric space* if it is equipped with a notion of *metric distance*, i.e. a function $d : X \times X \rightarrow \mathbb{R}$ such that, for any three elements x, y and z in X , which are called *points*, it holds:

1. $d(x, y) \geq 0$, i.e. it is *non-negative*.
2. $d(x, y) = 0$ implies $x = y$, i.e. it is *positive definite*.
3. $d(x, y) = d(y, x)$, i.e. it is *symmetric*.
4. $d(x, z) \leq d(x, y) + d(y, z)$, i.e. it satisfies the *triangle inequality*.

Example: The set $\mathbb{R}^n$ with the euclidean distance $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$ is a metric space.

Remark: There are metric spaces that are not finite-dimensional.

A. Open ball

The set

$$B_r(x) = \{y \in X : d(x, y) < r\}$$

is called the *open ball* of X centered at x and of radius $r > 0$.

B. Open set

A subset U of X is *open* if for every $x \in U$ there exists an open ball $B_r(x) \subset U$.

C. Complement set

Consider two subsets Y and Z of X . The *complement* $Y \setminus Z$ of Y in Z is the subset of X given by

$$Y \setminus Z := \{y \in Y : y \notin Z\}.$$

In other words, by the elements in Y that are not in Z . In particular, $Y^c = X \setminus Y$ is the subset of elements in X that are not in Y .

D. Closed set

A subset F of X is *closed* if its complement $X \setminus F$ is open.

E. Limit or accumulation point

A point x is a *limit point* (or *accumulation point*) of a subset A of X if, for every $r > 0$, it holds that

$$(B_r(x) \setminus \{x\}) \cap A \neq \emptyset.$$

Informally, this means that there are points in A that are different but arbitrarily close to x .

Example: The points 0 and 1 are limit points of the open interval $(0, 1)$.

F. Closure set

The *closure* $\overline{A}$ of a subset A of X is defined by

$$\overline{A} = A \cup \{\text{all limit points of } A\}.$$

Examples: the closed interval $[0, 1]$ is the closure of itself and the subsets $(0, 1)$, $[0, 1)$, $(0, 1]$ and $(0, 1/2) \cup (1/2, 1)$.

G. Bounded set

A subset A of X is *bounded* if there exists an $x \in X$ and a finite $r > 0$ such that $A \subset B_r(x)$.

H. Convergence

We say that a sequence $\{x_j \in X : j \in \mathbb{N}\}$ *converges* to a value $x \in X$ if, for each $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_j, x) < \epsilon$ for every $j \geq N$. It can be easily shown that, when the sequence is convergent, then the value of x is unique.

The convergence of the sequence is abbreviated by $x_j \rightarrow x$ as $j \rightarrow \infty$.

I. Cauchy sequence

A sequence $\{x_j \in X : j \in \mathbb{N}\}$ is a *Cauchy sequence* if, for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_i, x_j) < \epsilon$ for every $i, j \geq N$.

J. Complete metric space

A metric space X equipped with a metric d is called *complete* if every Cauchy sequence in it converges to an element in it. More formally, if $\{x_j \in X : j \in \mathbb{N}\}$ is a Cauchy sequence, then there exists an $x \in X$ such that, for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that $d(x_j, x) < \epsilon$ for every $j \geq N$.

Example: Consider the sequence given by

$$x_j = \sum_{i=0}^j \frac{1}{i!}.$$

Clearly, $x_j \rightarrow e$ as $j \rightarrow \infty$, i.e. it converges to Euler's number. This is a Cauchy sequence in $\mathbb{Q}$ but it does not converge to a value in $\mathbb{Q}$ because e is irrational. It converges to a value in $\mathbb{R}$. Therefore, $\mathbb{Q}$ is not a complete space.

II. NORMED VECTOR SPACE

A *normed vector space* V over a field $\mathbb{F}$ is a vector space equipped with a norm, which is a mapping $V \ni v \rightarrow |v| \in \mathbb{R}$ that, for any $v, w \in V$ and $\alpha \in \mathbb{F}$, it satisfies:

1. triangle inequality, $|v + w| \leq |v| + |w|$,
2. absolute homogeneity, $|\alpha v| = |\alpha||v|$, and
3. positive definiteness, $|v| = 0$ implies $v = 0$.

In particular, a *semi-norm* is a norm without property 3.

A. A normed vector space is a metric space

A normed vector space is a metric space because a norm defines a metric distance

$$d(v, w) = |v - w|.$$

To see this, it is enough to check that it actually satisfies the triangle inequality, because the remaining metric properties follow directly from the norm axioms. Namely,

$$d(v, w) = |v - w| = |v - u + u - w| \leq |v - u| + |u - w| = d(v, u) + d(u, w)$$

holds for any $v, u, w \in V$ due to property 1 of the norm.

B. Banach space

A complete normed vector space is called a *Banach space*. In other words, as we already saw, a normed vector space is a metric space. If additionally, such metric space is complete, meaning that all Cauchy sequences in it converge to vectors in it, then it is called a Banach space.

C. Continuous function

Consider two metric spaces X and Y . A function $f : X \rightarrow Y$ is continuous at $x \in X$ if, for every convergent sequence $x_j \rightarrow x$ it holds that $f(x_j) \rightarrow f(x)$.

III. HILBERT SPACE

A. Inner product space

Given a vector space V over $\mathbb{C}$, an *hermitian inner product* is a function $V \times V \ni (v, w) \rightarrow v \cdot w \in \mathbb{C}$ satisfying, for each $v, w, u \in V$ and $\alpha, \beta \in \mathbb{C}$, the following properties:

1. linearity of the second argument, $v \cdot (\alpha w + \beta u) = \alpha(v \cdot w) + \beta(v \cdot u)$,
2. conjugate symmetry, $v \cdot w = (w \cdot v)^*$, and
3. positive definiteness, $v \cdot v \geq 0$ and $v \cdot v = 0$ only if $v = 0$.

A vector space equipped with an (hermitian) inner product is called an inner product space.

Remark: We adopt the physics convention where the inner product is linear in the second argument and antilinear in the first.

B. Induced norm

The mapping $V \ni v \rightarrow \sqrt{v \cdot v}$ is a norm called the *induced norm*. Hence, an inner product space is a normed space.

C. Cauchy-Schwartz inequality

An hermitian inner product satisfies the so called Cauchy-Schwartz inequality

$$|v \cdot w| \leq |v||w|$$

which holds for all $v, w \in V$.

Excercise: Prove the Cauchy-Schwartz inequality.

D. Hilbert space

A Hilbert space is a complete hermitian inner product space. In other words, in a Hilbert space H , all Cauchy sequences converge to a vector in H , where the induced norm is used to determine convergence.

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